

The whole course

11 units across OCR GCSE Computer Science (9-1) J277, and the Python pathway beyond it

The qualification

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component 01	Computer systems — 80 marks, 1 hour 30 minutes
Component 02	Computational thinking, algorithms and programming — 80 marks, 1 hour 30 minutes
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

The 12 units

TOPIC	UNIT	PAPER	LESSONS	ACTIVITIES	MARKS	OUTCOMES	COVERAGE
1.1	Systems architecture	Component 01	5	50	82	12	11 / 1 / 0
1.2	Memory and storage	Component 01	18	221	316	41	32 / 6 / 3
1.3	Computer networks, connections and protocols	Component 01	15	175	277	39	28 / 10 / 1
1.4	Network security	Component 01	12	122	199	20	20 / 0 / 0
1.5	Systems software	Component 01	5	55	103	10	10 / 0 / 0
1.6	Ethical, legal, cultural and environmental impacts of digital technology	Component 01	9	92	197	19	18 / 1 / 0
2.1	Algorithms	Component 02	13	155	361	31	31 / 0 / 0
2.2	Programming fundamentals	Component 02	8	99	135	21	18 / 0 / 3
2.3	Producing robust programs	Component 02	7	85	129	18	14 / 4 / 0
2.4	Boolean logic	Component 02	5	41	97	5	5 / 0 / 0
2.5	Programming languages and Integrated Development Environments	Component 02	5	70	128	15	11 / 2 / 2

TOPIC	UNIT	PAPER	LESSONS	ACTIVITIES	MARKS	OUTCOMES	COVERAGE
PY	Python programming	beyond J277	13	627	1242	39	32 / 7 / 0

115 lessons, 1792 activities, 3266 marks and 270 stated outcomes in all. Coverage reads **full** / **partial** / **not covered**, and means taught, practised and assessed **within that unit**: 230 / 31 / 9.

Which units sit in which paper

Component 01: Computer systems

1.1 Systems architecture, **1.2** Memory and storage, **1.3** Computer networks, connections and protocols, **1.4** Network security, **1.5** Systems software, **1.6** Ethical, legal, cultural and environmental impacts of digital technology

Component 02: Computational thinking, algorithms and programming

2.1 Algorithms, **2.2** Programming fundamentals, **2.3** Producing robust programs, **2.4** Boolean logic, **2.5** Programming languages and Integrated Development Environments

Beyond the specification

The Python course is not an OCR topic and carries no specification reference. It teaches programming in a real language, which Component 02 assesses the thinking behind: a pupil who has written, run and fixed their own programs reads an exam question about iteration differently from one who has only read about it. Where it supports a topic, that topic is named. Where it goes beyond the specification - the editor, the test-based marking, SQL through sqlite3 - it is named as enrichment, not as coverage.

What is not covered here

Every topic this course names has a unit. That is a statement about this course's own topic list, not a claim that the specification contains no other requirement: the specification document was not available to check against when this was built.

Coverage is not attainment. A full row says the course teaches, practises and assesses an outcome somewhere. It says nothing about whether a particular pupil has learnt it. Only your own assessment does that.

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